

Grade 6 Accelerated
Unit 6
Lesson 1 Practice

Name _____
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Complete the table by rewriting the numeric expression in the form indicated. Then, find the value of each expression.

	Exponential Form	Expanded Form	Value
1.	$(3.2)^3$		
2.		$(9)(9)(9)$	
3.	$\left(\frac{3}{5}\right)^4$		
4.		$\frac{5}{2} \cdot \frac{5}{2} \cdot \frac{5}{2} \cdot \frac{5}{2} \cdot \frac{5}{2}$	

5. What is the value of the expression $(2.8)^2$?

6. Which expression has the larger value, 3^6 or 6^3 ? Show your work or explain your reasoning.

7. Select all expressions that are equivalent to 5^4 .

☐ $5 + 5 + 5 + 5$

☐ 20

☐ $5 \cdot 5 \cdot 5 \cdot 5$

☐ 625

☐ $4 \cdot 4 \cdot 4 \cdot 4 \cdot 4$

8. Select all expressions that are equivalent to $\left(\frac{1}{7}\right)^3$.

☐ $\frac{3}{7}$

☐ 512

☐ $\frac{3}{21}$

☐ $\frac{1}{7} \cdot \frac{1}{7} \cdot \frac{1}{7}$

☐ $\frac{1}{343}$

9. What is the value of the expression $(0.08)^3$?

10. What is the value of the expression $(4.9)^2$?

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Lesson 2 Practice

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Find the prime factorization of each value.

1. 275

2. 160

3. 232

4. 54

5. 2550

6. 1020

7. 45

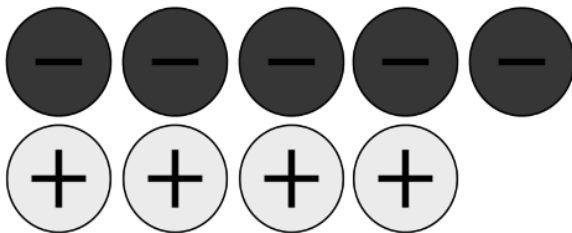
8. 216

9. Find a number that meets the following conditions.
- The value of the number is less than 500.
 - One of its prime factors has an exponent of 5.
 - When expressed in exponential form, its prime factorization has 3 bases.
10. Find a number that meets the following conditions.
- The value of the number is less than 5000.
 - One of its prime factors has an exponent of 2.
 - When expressed in exponential form, its prime factorization has 5 bases.

1. Which situation could be represented by the numerical expression $-8 + 12$?

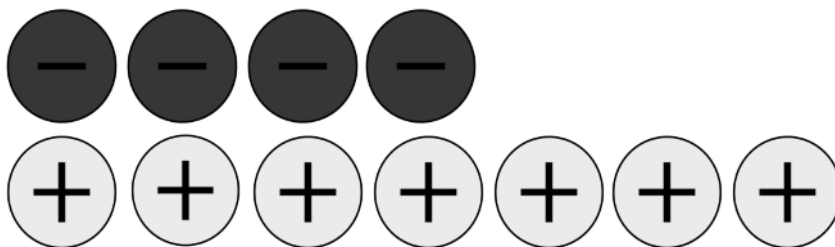
- (A) Jake deposited \$4 into his account and withdrew \$12.
- (B) Benjamin withdrew \$8 from his account and withdrew another \$12.
- (C) Carol deposited \$8 into her account and withdrew \$10.
- (D) Debbie withdrew \$8 from her account and deposited \$12.

2. Saoirse created the sketch below to represent the expression $-5 + (-4)$.



Her sketch is incorrect. Explain why, and create a new sketch to find the sum.

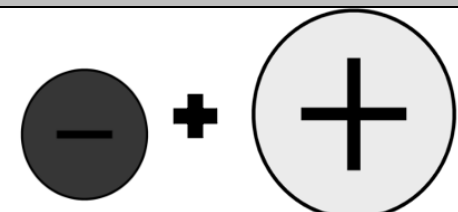
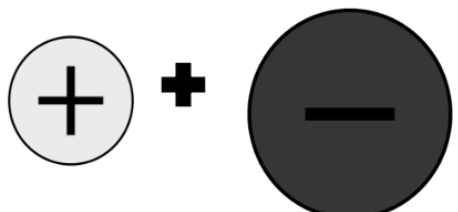
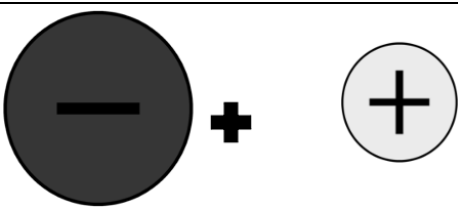
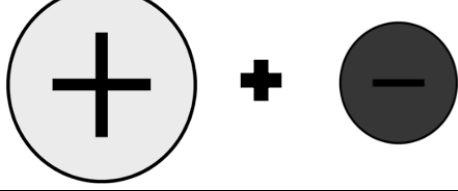
3. Integer chips are shown.



Which scenario could be represented by the integer chips?

- (A) Octavian deposited \$4 into his account and withdrew \$3.
- (B) Thelonious withdrew \$4 from his account and withdrew another \$7.
- (C) Lauren deposited \$4 into her account and withdrew \$7.
- (D) Linda withdrew \$4 from her account and deposited \$7.

Write an example of the rule. The first row has been completed.

	Visual Model of Addition	Sign of Sum	Example
			$-3 + 7 = 4$
4.			
5.			
6.			

Complete the table for the expression below.

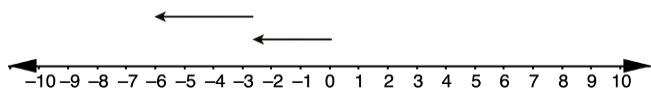
	Expression	$4 + ^{-}3$
7.	Integer Chips	
8.	Money Sentence	
9.	Sign of the Sum	
10.	Sum	

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Lesson 4 Practice

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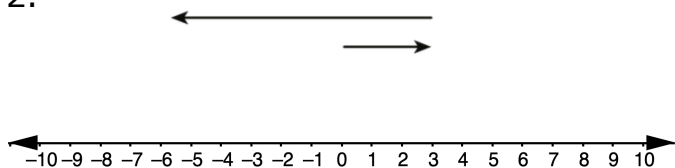
For each number line model, write the addition expression represented and find the sum.

1.



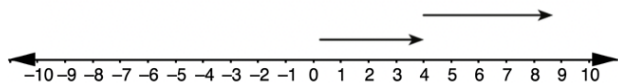
Addition Expression	Sum

2.



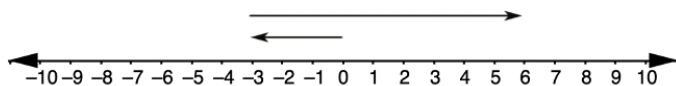
Addition Expression	Sum

3.



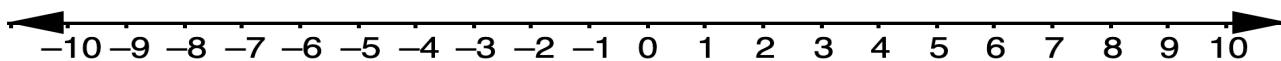
Addition Expression	Sum

4.



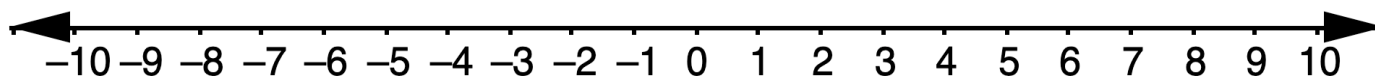
Addition Expression	Sum

5. Use the number line to show the sum of $-1 + 5$. Then, complete the equation.



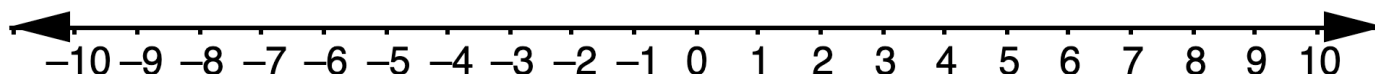
$$-1 + 5 = \underline{\hspace{2cm}}$$

6. Use the number line to show the sum of $-2 + (-6)$. Then, complete the equation.



$$-2 + (-6) = \underline{\hspace{2cm}}$$

7. Use the number line to show the sum of $-9 + 3$. Then, complete the equation.



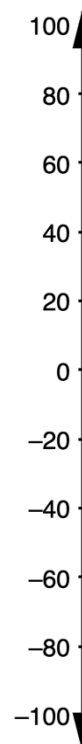
$$-9 + 3 = \underline{\hspace{2cm}}$$

In a single day, the temperature in degrees Fahrenheit ($^{\circ}\text{F}$) of a town in Canada began at a low of -60°F and increased 20°F by noon.

8. Plot a point at the low temperature.

9. Draw a ray along the number line to represent the increase of 20°F .

10. What was the temperature in the town by noon?



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Lesson 5 Practice

Name _____
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Find each sum.

1. $317 + (-544)$

2. $-756 + 284$

3. $3,579 + (-1,987)$

4. $-5,000 + (-1)$

5. $-86 + (-229)$

6. $-8,352 + 4,686$

7. $-4,650 + 0$

8. $3,540 + (-3,540)$

9. $-234 + 689$

10. $-92,547 + 2,113$

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Lesson 6 Practice

Name _____
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Use integer chips to find the value of each expression.

1. $-1 - 2$

2. $0 - (-3)$

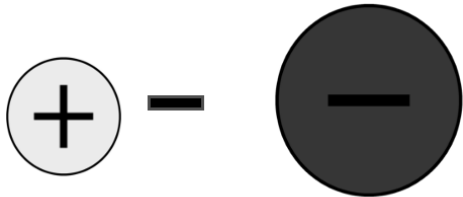
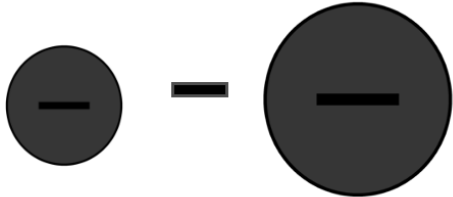
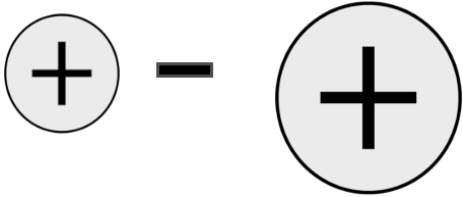
3. $3 - (-4)$

4. $5 - 4$

5. $2 - 5$

6. $6 - 11$

Complete the table.

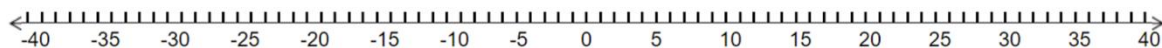
	Visual Model of Subtraction	Sign of Sum	Example
7.			
8.			$-3 - 6 = -9$
9.			
10.			$2 - 8 = -6$

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Lesson 7 Practice

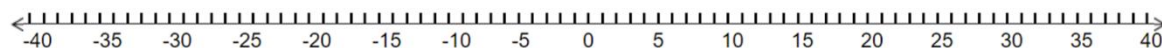
Name _____
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In questions 1-6, represent each statement on a number line and find the value.

1. $-26 - (-28)$



2. $23 - 30$



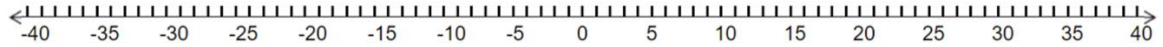
3. $-1 - 15$



4. $(-13) - (-7)$



5. $(-1) - 18$



6. $29 - 11$



In questions 7-10, rewrite each subtraction expression as an addition expression and find the value.

7. $26 - 32$

8. $-43 - 18$

9. $-15 - (-43)$

10. $(-8) - (-16)$

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Lesson 8 Practice

Name _____
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Evaluate each expression.

1. $-233 - 348$

2. $-409 - (-176)$

3. $503 - 568$

4. $355 - (-213)$

5. $622 - 545$

6. $712 - (-713)$

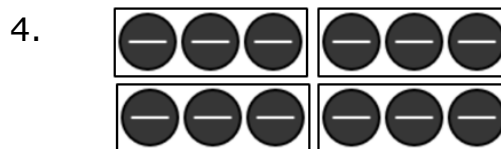
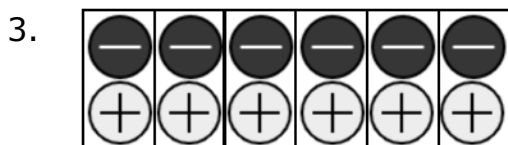
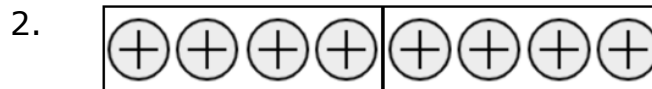
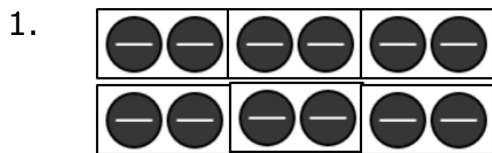
7. $-188 - 356$

8. $419 - 455$

9. $-366 - (-402)$

10. $-523 - (-623)$

For questions 1-6, write an expression to represent the multiplication model and provide the value.



For questions 5-6, create a model of the expression and provide the value.

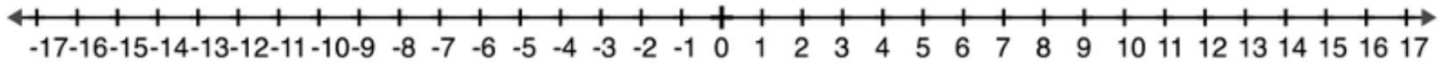
5. $(-5) \times 3$

Model	Value

6. $6(-7)$

Model	Value

For questions 7-8, use the number line to evaluate the expression.



7. $-4 \cdot 3$

8. $(-6) \times (-2)$

For questions 9-10, evaluate the expression.

9. $(-11)(-9)$

10. $9(-4)$

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Lesson 10 Practice

Name _____
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For questions 1-6, determine the product or quotient.

1. $-28 \div 4$

2. $(56) \div (-7)$

3. $\frac{-48}{-6}$

4. $(-65) \div (-5)$

5. $\frac{24}{-8}$

6. $-48 \div 2$

For questions 7-10, create two equivalent expressions for each division expression and provide the value.

7. $(-66) \div 6$

8. $\frac{32}{-2}$

9. $-60 \div (-5)$

10. $64 \div (-8)$

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Lesson 11 Practice

Name _____
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Determine the product or quotient.

1. $-52 \div 4$

2. $(17) \cdot (-9)$

3. $\frac{-112}{7}$

4. $(-85) \div (-5)$

5. $(-6)(-13)$

6. $\frac{-111}{-3}$

7. $(-6) \times (-19)$

8. $8 \cdot (-23)$

9. $(-9)(14)$

10. $132 \div (-4)$

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Lesson 12 Practice

Name _____
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Determine the product or quotient.

1. $-368 \div 16$

2. $(48) \cdot (-18)$

3. $\frac{-255}{17}$

4. $(-192) \div (-8)$

5. $(-1) \cdot (34) \cdot (-2)$

6. $\frac{-224}{-4}$

7. $(-17) \times (-18) \times (1)$

8. $5 \cdot (-12) \cdot 6$

9. $(-9) \cdot (-8) \cdot (-2)$

10. $108 \div (-12)$

Grade 6 Accelerated
Unit 6
Lesson 13 Practice

Name _____
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Evaluate each expression.

1. $(3)^5$

2. $(-3)^3$

3. $\left(-\frac{2}{3}\right)^4$

4. $\left(\frac{1}{3}\right)^5$

5. The expression $(-6)^3$ will result in a

- ☐ negative
- ☐ positive

value because

the

- ☐ base
- ☐ exponent

is an

- ☐ odd
- ☐ even

number.

For questions 6-10, select all equivalent values or expressions.

6. $(1.8)^4$

- ☐ 7.2
- ☐ $(4)(1.8)$
- ☐ 10.4976
- ☐ $(3.24)(3.24)$
- ☐ $1.8 \cdot 1.8 \cdot 1.8 \cdot 1.8$

7. $(-3)^5$

- ☐ -15
- ☐ -243
- ☐ $(5)(-3)$
- ☐ $(9)(-3)$
- ☐ $(-3)(-3)(-3)$

8. $\left(\frac{1}{4}\right)^3$

- ☐ $\frac{1}{12}$
- ☐ $\frac{1}{64}$
- ☐ $\left(\frac{1}{16}\right)(3)$
- ☐ $\left(\frac{1}{16}\right)\left(\frac{1}{4}\right)$
- ☐ $\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4}$

9. $(-5)^4$

- ☐ 625
- ☐ $(25)^2$
- ☐ -625
- ☐ $(-25)(-25)$
- ☐ $(-5)(-5)(-5)(-5)$

10. $(-9)(-9)(-9)$

- ☐ $(9)^3$
- ☐ 729
- ☐ $(-9)^3$
- ☐ -729
- ☐ $-9 \cdot (-9)^2$